

AI as a Thinking Partner

Why the usual rollout yields faster workers with weaker judgment, and what two trials show instead

Marco Policani

Enterprise Portfolio · PMO · AI Operating Governance

policani.net · linkedin.com/in/marcpolicani

March 2026

EXECUTIVE SUMMARY

Before AI challenges the analysis, make the person state a view, the key assumptions, and the evidence that could change it.

A team can produce AI-assisted work faster while independent judgment becomes harder to see. Two student trials show why the way people use the tool matters, but they do not prove the same effect in a company.

- 01** In one trial, answer-first AI left students 17 percent below a no-AI group on a later exam. A guarded version of the same model largely avoided that loss.
- 02** For high-judgment work, use a short pre-mortem: state the first view, name the assumptions, and say what evidence would change it before AI responds.
- 03** Support the habit through training, safe work rules, and managers who ask who checked the evidence and who makes the final decision.

CONTENTS

When faster output hides weaker judgment	A better standard: AI as a thinking partner
The design gap that defines the outcome	Two tracks, not one timeline
What the workplace evidence adds	The three-layer change model
Why the usual responses fail	What to do next
Why expertise becomes more valuable	Notes and sources

When faster output hides weaker judgment

You lead an AI program with rising use and faster output. Yet one question remains open: Are people getting better at judging the work, or only faster at producing it? That difference matters when the answer is wrong, the case is unusual, or the tool is unavailable.

A preliminary MIT NANDA report estimated that enterprises had invested \$30–40 billion in generative AI while 95 percent of the organizations it studied reported no measurable return. The report does not prove that weak judgment caused the gap. It does show why leaders need a stronger test than adoption or output volume. [1]

Two randomized trials show how much the way people use AI can matter. In one, students with an answer-first AI tool later scored 17 percent worse on an exam without AI than students who had no AI. A guarded

version of the same model made students work through hints and largely avoided that loss. In another trial, a custom AI tutor produced more than twice the median learning gain of a class using strong active-learning methods. [4] [19]

These studies involved students, not experienced workers. They do not prove the same effect in a company. They do show that access to a capable model does not settle the outcome. The interaction can either keep reasoning active or let people bypass it.

This is already an operating problem. A Microsoft and LinkedIn survey found that 78 percent of AI users brought their own tools to work, while only 39 percent had received company training. A separate survey reported by SANS found that about 48 percent would keep using AI even if their employer banned it. [2] [3] The habit often forms before the organization sets a standard.

The design gap that defines the outcome

Bastani and colleagues studied nearly 1,000 high-school students. Classes received a standard AI tool, a guarded tutor, or no AI. The standard tool could give a direct answer. The guarded tutor used the same underlying model but led students through hints. On a later exam without AI, the standard-tool group scored 17 percent below the no-AI group. The guarded group was near the no-AI group, though not above it. [4]

EXHIBIT 1

The way people use AI changes the result

**-17% /
+2×**

Two separate student trials: answer-first AI left one group 17 percent below a no-AI group on a later exam; a custom reasoning-first tutor more than doubled median learning gains against active learning.

Bastani et al., PNAS 2025; Kestin et al., Scientific Reports 2025 (see Notes).

Kestin and colleagues studied introductory physics at Harvard. They compared a custom AI tutor with active learning that used peer work and instructor feedback. The AI group had more than twice the median learning gain and used a median of 49 minutes, compared with a 60-minute class. The study covered two topics, a selected student group, and only two weeks. The authors did not claim that the result would carry into complex professional judgment. [19]

The two trials used different students, subjects, and designs. They should not be treated as one direct comparison. Read together, they make a narrower point: an answer-first tool and a reasoning-first tool can lead to very different results. Leaders choose that pattern through product design, training, and work rules.

What the workplace evidence adds

Workplace studies show the same need for care, but they do not close the evidence gap. In a field experiment with 758 consultants, AI users completed more work, worked faster, and produced higher-rated results on tasks within the model's tested range. On one task outside that range, they were 19 percentage points less likely to reach the correct answer. People whose results fell tended to accept the AI output and question it less. [5]

Confidence can make this harder. One study found that several major language models were overconfident most of the time. Another study followed 257 medical students through 3,855 decisions. AI explanations helped when the advice was right and hurt when it was wrong, because both explanations sounded convincing. The most capable students lost the most when the advice was wrong. [6]

Other studies add warning signs, with important limits. A survey of 319 knowledge workers linked greater confidence in AI with less reported critical-thinking effort. It measured self-reports, not objective performance. [10] A study of 666 people linked frequent AI use with weaker critical-thinking scores through cognitive offloading, but it was correlational and did not test a better use pattern. [13] Four online experiments found that AI improved task results but that later solo work brought lower motivation and more boredom. [14]

An MIT Media Lab preprint used EEG data to study AI-assisted writing. Formal commentary raised concerns about its small later sample, methods, and reporting. It is directionally relevant, but it should not carry the recommendation. [12]

No large workplace trial has compared answer-first and reasoning-first AI for expert work. No study has directly compared trained workers with self-taught users who bring their own AI tools. Those gaps matter. The operating case in this paper is a bounded inference: when judgment matters, the work should require people to form, test, and own a view before they accept an answer.

Why the usual responses fail

A ban does not create a better habit. It can move use out of view. IBM reported that organizations with high levels of shadow AI had breach costs \$670,000 higher than those with little or none, and one in five reported a shadow-AI breach. Netskope reported an average of 223 monthly incidents in which users sent sensitive data to AI apps. An UpGuard survey reported that 90 percent of security leaders used unapproved tools themselves. Another survey found that 38 percent of respondents had shared confidential work data with AI without approval. [7] [8] [9] [15]

Those figures do not prove that every ban increases risk. They show that hidden use is already common. The BYOD era offers a structural analogy, not direct AI evidence: when people have a real need, control works better when the company provides a usable path. [11] Microsoft's 2025 work report also describes the limits of top-down change. Applying that point to established AI habits is the paper's inference, not a result the report tested. [16]

The other common response is a broad "human in the loop" rule. That phrase says nothing about the person's skill, authority, or task. A reviewer who cannot spot a weak assumption is not a control. A person who sees the work only after the answer is accepted is just a signature step.

Four failures sit underneath both the ban and the weak review step:

- The task is framed only as completion. The worker is rewarded for finishing, not for showing the reasoning.
- The organization measures use and output, but not the quality of the decision.
- The tool removes the productive effort that helps people build skill.
- No named person has to challenge the answer, add missing context, and own the conclusion.

The education system can reinforce the same habit before people reach work. A University of Florida policy brief describes how schools respond to AI use, while a secondary summary of UK and U.S. surveys reports fast growth in student use and uneven teacher preparation. These sources describe the setting; they do not prove later workplace behavior. [20] [21]

Why expertise becomes more valuable

Autor and Thompson offer a useful labor-economics frame. When technology removes less-skilled tasks from a job, the remaining expert work can become more valuable. When it removes expert work, wages can fall even as employment grows. They do not apply this claim directly to generative AI. This paper does: as AI handles more drafting, search, and routine synthesis, the value of sound judgment can rise. [22]

That judgment includes setting the goal, testing evidence, finding exceptions, and deciding when the model is outside its depth. Those are also the human tasks that matter more as systems take on longer chains of work. Faster output has value. It does not replace the need to know whether the output should be trusted.

A better standard: AI as a thinking partner

AI acts as a thinking partner when it helps a person surface, challenge, and test their own reasoning. The person still owns the conclusion, while the tool helps expose weak assumptions before the decision is made.

Use that standard for work with real judgment: strategy, large resource choices, client-facing analysis, and risk decisions. Routine, low-risk work can use more automation. When the model's limits are unknown or the cost of error is high, treat the work as outside the safe range until someone proves otherwise.

Start with a short pre-mortem

Before using AI on high-judgment work, ask the worker to do three things without it:

- State the answer or recommendation they expect, and why.
- Name the two or three assumptions that must be true.
- Say what evidence would change their mind.

Then use AI to challenge that first view. Ask it to find the weakest assumption, raise the best counterargument, and point to missing evidence. The worker checks those challenges against reliable

sources and decides what survives.

This adapts Gary Klein's team pre-mortem to individual AI-assisted work. A controlled test found that the original pre-mortem reduced overconfidence in plans more than critique or pro/con lists. The AI adaptation has not been validated in that exact form. [17]

Related evidence is promising but limited. Ma and colleagues tested a “think the opposite” prompt without AI, then tested confidence calibration in an AI-assisted task. Calibration reduced under-reliance and improved performance, but it did not reduce over-reliance. [18] The protocol changes the worker's posture. It does not make the model well calibrated.

Two tracks, not one timeline

Some workers still have strong independent judgment and need to protect it. Others may have lost practice after long use of answer-first tools. Those are different problems.

EXHIBIT 3

Protecting skill and rebuilding it are different jobs

Track 1 — protect current judgment

Test the pre-mortem and real review step over 60–90 days.

Track 2 — rebuild lost practice

Use AI-free work and expert feedback. The longer timeline is a planning assumption, not a validated program result.

- Track 1 — protect. Install the pre-mortem habit and a real review step over 60–90 days. The goal is to keep current skill active.
- Track 2 — restore. Rebuild independent reasoning through work done without AI, expert feedback, and repeated practice. This will take longer.

The 60–90 day and 12–36 month ranges are planning assumptions, not findings from a validated program. A proven restorative curriculum for experienced workers does not yet exist. Leaders should say so, test carefully, and avoid promising a fast recovery.

Start the assessment with two inputs. First, learn which tools people use and for which tasks. Separate honest disclosure from discipline. Second, ask people to complete job-relevant judgment tasks without AI. Review the reasoning, not just the answer. That shows whether training should protect a skill or help rebuild it.

The three-layer change model

The change must reach training, the way work is set up, and management review at the same time. If one layer is missing, deadline pressure will pull people back to answer-first use.

EXHIBIT 2**Support the habit in three places**

- 1 Training design — CHRO and L&D**
Assess current habits, teach the pre-mortem, explain the tool's limits, and score reasoning rather than output.
- 2 Usage configuration — IT Security and Operations**
Provide a safe tool, require the pre-mortem for high-judgment work, and flag tasks handed fully to AI.
- 3 Institutional accountability — managers and executives**
Ask what changed the worker's view, name who owns the decision, reward sound reasoning, and protect the pilot from a speed-only scorecard.

Layer 1 — training design

The CHRO and learning team own the training. Frame it as better decision quality, not as tool adoption.

- Assess current habits before teaching one course to everyone.
- Show where the tool works well and where people must slow down.
- Teach people to test assumptions and evidence, not just write prompts.
- Score the reasoning behind a decision, not only the finished output.

Layer 2 — work setup

IT, security, and operations own the approved path. Give people a useful, safe option instead of relying on a ban.

- Require the pre-mortem for high-judgment tasks.
- Flag work that people hand fully to AI without an independent check.
- Match the review step to the cost of error and the model's known limits.
- Keep confidential data inside approved tools and access rules.

Layer 3 — management accountability

Managers and executives make the standard real.

- Ask what the person believed before using AI and what changed their view.
- Name who checked the evidence and who makes the final decision.
- Add reasoning quality to reviews, promotion choices, and project lessons.
- Protect the first pilot from a scorecard that rewards speed alone.

In my own portfolio and delivery work, the rule was simple: AI proposes; named people check, change, or reject. AI could flag schedule conflicts or suggest changes across many records. It did not make the

decision. Product and delivery owners accepted, changed, or rejected each proposal. This kept judgment visible where the choice carried consequences without turning review into needless delay.

What to do next

Choose one function where reasoning quality already matters, such as strategy, risk, client advice, or product decisions. Name one executive owner. Before the pilot starts, define how reviewers will judge the reasoning and the result.

For one quarter, require the three-step pre-mortem before people use AI on selected high-judgment work. Record the first view, the assumptions, the evidence that could change it, the AI challenge, and the final human decision. Compare speed, rework, error, and review quality with the old way of working.

At the end of the quarter, decide whether to expand, change, or stop. Do not claim that a short pilot has restored lost skill. It can show whether a reasoning-first pattern fits the work and whether managers will support it.

The core choice is plain. An answer-first system can make output easier while leaving judgment weaker or hidden. A thinking-partner system makes the worker form a view, test it, and own the conclusion. The model may be the same. The way people work with it is not.

Notes and sources

[1] Challapally, A., Pease, C., Raskar, R., and Chari, P. (2025). The GenAI Divide: State of AI in Business 2025 (v0.1). MIT NANDA. The \$30–40B enterprise investment figure and the 95% zero-return figure are from this preliminary report. Methodology: 52 structured interviews, 300+ disclosed AI initiatives, 153 senior leaders. https://mlq.ai/media/quarterly_decks/v0.1_State_of_AI_in_Business_2025_Report.pdf

[2] Microsoft and LinkedIn, 2024 Work Trend Index Annual Report. Survey of 31,000 people, 31 markets, Feb–Mar 2024. The 78% BYOAI figure includes any personal tool use supplementary to employer-provided tools and may overstate fully unsanctioned behavior. Only 39% of AI users had received AI training from their company. <https://www.microsoft.com/en-us/worklab/work-trend-index/ai-at-work-is-here-now-comes-the-hard-part>

[3] SANS Institute, "Sunlight AI: Bringing Shadow AI Into the Light," December 2025. The 48% would-not-stop figure draws on October 2024 survey data. <https://www.sans.org/blog/sunlight-ai-bringing-shadow-ai-light>

[4] Bastani, H., Bastani, O., Sungu, A., Ge, H., Kabakci, O., and Mariman, R. (2025). Generative AI without guardrails can harm learning. PNAS, 122(26), e2422633122. Pre-registered RCT; ~1,000 students, 9th–11th grade; Turkey; IRB-approved. August 2025 correction updated one author's affiliation only; findings unchanged. <https://doi.org/10.1073/pnas.2422633122>

[5] Dell'Acqua, F., et al. (2026). Navigating the jagged technological frontier. Organization Science, 37(2), 403–423; first circulated as HBS Working Paper No. 24-013 (2023). Pre-registered; n=758 BCG consultants. "Blindly adopt" wording is from the working paper; its retainment measure (copying AI output) was associated with better in-frontier performance. BCG co-designed a study finding BCG consultants

perform better with AI; pre-registration and multi-institutional authorship partially mitigate this. <https://www.hbs.edu/faculty/Pages/item.aspx?num=64700>

[6] Khanna, M., Wang, Z., Wei, L., and Xue, L. (2025). When Medical AI Explanations Help and When They Harm. arXiv:2512.08424. n=257 medical students; 3,855 diagnostic decisions. <https://arxiv.org/abs/2512.08424>. Also: Groot, T., and Valdenegro-Toro, M. (2024). Overconfidence Is Key: Verbalized Uncertainty Evaluation in GPT-4, GPT-3.5, LLaMA2 and PaLM 2. Proceedings of TrustNLP 2024, ACL. <https://aclanthology.org/2024.trustnlp-1.13/>

[7] IBM, Cost of a Data Breach Report 2025 (Ponemon Institute), July 30, 2025. Organizations with high shadow-AI levels saw \$670,000 higher breach costs than those with low or no shadow AI; one in five organizations reported a breach due to shadow AI. <https://www.ibm.com/security/data-breach>

[8] Netskope, Cloud and Threat Report, October 2024–October 2025. 223 incidents a month of users sending sensitive data to AI apps, as reported by Cybersecurity Dive; the count doubled year over year. <https://www.cybersecuritydive.com/news/shadow-ai-security-risks-netskope/808860/>

[9] UpGuard survey of security leaders, 2025. Reported in Fortra, November 2025. <https://www.fortra.com/blog/shadow-ai-security-breaches-will-hit-40-companies-2030-warns-gartner>

[10] Lee, H-P., et al. (2025). The impact of generative AI on critical thinking. CHI '25, Yokohama. Microsoft Research. "Task stewardship" is the authors' own framing. Limitations: self-report; cross-sectional; critical thinking not objectively measured; causality not established. <https://www.microsoft.com/en-us/research/publication/the-impact-of-generative-ai-on-critical-thinking>

[11] BYOD-era security practice is cited only as a structural analog; no AI-specific provision study of equivalent design currently exists.

[12] Kosmyrna, N., et al. (2025). Your brain on ChatGPT. arXiv:2506.08872. MIT Media Lab. Preprint — not peer-reviewed as of March 2026. Formal commentary arXiv:2601.00856 identifies five methodological concerns. Cited as directionally consistent only. <https://arxiv.org/abs/2506.08872>

[13] Gerlich, M. (2025). AI tools in society: Impacts on cognitive offloading and the future of critical thinking. Societies, 15(1), 6. n=666. Correlational; recommends balanced AI integration, which the study did not test. No objective performance measures. <https://doi.org/10.3390/soc15010006>

[14] Wu, S., Liu, Y., Ruan, M., Chen, S., and Xie, X.-Y. (2025). Human-generative AI collaboration enhances task performance but undermines human's intrinsic motivation. Scientific Reports. Four online experiments (Prolific); N=3,562. Yukun Liu is corresponding author. <https://www.nature.com/articles/s41598-025-98385-2>

[15] CybSafe and National Cybersecurity Alliance, survey of 7,000 respondents, late 2024. The 38% confidential-data-sharing figure is from Cloud Security Alliance analysis. <https://cloudsecurityalliance.org/blog/2025/03/04/ai-gone-wild-why-shadow-ai-is-your-it-team-s-worst-nightmare>

[16] Microsoft, New Future of Work Report 2025. The inference that resistance to top-down mandates extends specifically to established BYOAI interaction habits is the author's extension; not directly established by the cited source. <https://www.microsoft.com/en-us/research/wp-content/uploads/2025/12/New-Future-Of-Work-Report-2025.pdf>

- [17] Klein, G. (2007). Performing a project premortem. *Harvard Business Review*, 85(9), 18–19. Validated in: Veinott, E. S., Klein, G., and Wiggins, S. (2010). Evaluating the effectiveness of the PreMortem technique on plan confidence. ISCRAM, Seattle. The adaptation here — individual cognitive priming before AI engagement — extends Klein's team-based technique and has not been independently validated in that specific form. <https://hbr.org/2007/09/performing-a-project-premortem>
- [18] Ma, S., et al. (2024). "Are you really sure?" CHI '24, Honolulu. Three studies; "Think the Opposite" was tested in Study 2 without AI assistance, and Study 3 found that confidence calibration reduced under-reliance on AI but not over-reliance. The intervention is similar to the pre-mortem prompt protocol and was developed independently. Limitation: income-prediction task in a lab setting. <https://dl.acm.org/doi/10.1145/3613904.3642671>
- [19] Kestin, G., Miller, K., Kiales, A., Milbourne, T., and Ponti, G. (2025). AI tutoring outperforms in-class active learning. *Scientific Reports*. RCT with crossover design; 233 enrolled, 194 eligible Harvard undergraduates; introductory physics; fall 2023. Effect size 0.73–1.3; $p < 10^{-8}$. Critical limitations: selected population; two topics; introductory material; authors do not presume the result extends to complex synthesis and higher-order critical thinking. <https://www.nature.com/articles/s41598-025-97652-6>
- [20] Curran, F. C., and Goo, M. (2025). Disciplining AI Use. University of Florida Education Policy Research Center policy brief, reporting K-12 teacher survey findings collected by Dwyer, L., and Laird, E. (2024). https://education.ufl.edu/eprc/files/2025/11/EPRC_PolicyBrief-AI-and-School-Discipline-Curran-and-Goo-2025.pdf
- [21] HEPI, Student Academic Experience Survey 2025. UK university student AI use for assessments: 53% (2024) to 88% (2025). HEPI 2025 also reports 67% saying good GenAI skills are essential. The 38% teacher figure is from the Walton Family Foundation / Impact Research teacher survey (February 2023). <https://www.demandsage.com/ai-in-education-statistics/>
- [22] Autor, D., and Thompson, N. (2025). Expertise. NBER Working Paper 33941. The inference that AI is specifically automating the non-expert tasks of knowledge work is the author's application of the framework; Autor and Thompson do not make this specific claim about generative AI. <https://www.nber.org/papers/w33941>

About the author

Marco Policani is an enterprise portfolio, PMO, and AI operating-governance leader. He builds portfolio governance systems where AI carries the analytical load and named humans own every decision — an approach documented across case studies, walkthroughs, and working governance guides on his portfolio site.

Portfolio & governance library: policani.net/governance · Full portfolio: policani.net · LinkedIn: linkedin.com/in/marcpolicani

This white paper may be shared freely with attribution. © 2026 Marco Policani.